Negative Hours

Predicting negative power prices

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1 Problem statement

Electricity for tomorrow is sold in a daily auction that closes at 12:00 and sets one price per hour. When a lot of sun or wind meets low demand, e.g. a sunny spring Sunday at noon, more power is produced than anyone needs, and many plants cannot or will not switch off. The price then falls below zero: producers pay to deliver, and consumers are paid to take power. This is no longer rare: the German-Luxembourg market (DE-LU) had 389 negative hours in the first half of 2025 alone. Everyone has to bid before knowing whether it will happen. NegativeHours predicts, at 10:00 on the day before, which hours of tomorrow will go negative.

Who benefits	Decides before 12:00	Economic benefit	Resource benefit
Solar & wind marketers	below which price each plant is switched off	no paying to deliver, no support payment lost	power is not produced when nobody needs it
Battery & storage operators	when to charge	paid to charge, then sell in the evening peak	surplus solar is stored instead of turned off, less gas power in the evening

PREDICT	HORIZON	SCOPE
P(price < 0 €/MWh) + price, for each of the 24 h of D	**Issued 10:00 on D-1** 2 h before auction gate closure	**DE-LU bidding zone** hourly · 2024 to today

Success criterion: PR-AUC ≥ 0.75 on the test year and an F1 score at least 0.10 higher than the best baseline.
Baselines: naïve yesterday (same hour on D-1) and weekly naïve (same hour on D-7).
Notation: D is the delivery day being predicted; D-1 is the day before. When the forecast is made and the auction runs. D-2 and D-7 mean two days and one week before D. Example: Thursday 10:00 (D-1) forecasts Friday (D).

2 Originality & motivation

Negative prices are a symptom of an energy system in transition. Solar and wind capacity is growing faster than the flexibility to absorb it. On sunny, windy hours with low demand the market is flooded with power that nobody needs. Today much of that surplus is either turned off or sold at a loss, while a few hours later gas plants still have to cover the evening peak. Knowing in advance when this will happen lets producers avoid losses and lets storage operators take in clean energy that would otherwise be wasted. The topic becomes more relevant every year. Germany records more negative hours than ever, and Switzerland saw almost as many in 2024. I see it as a great MLOPS project. A real decision with a daily deadline, public data that is updated every day, and a forecast that can be checked against reality only a few hours later.

3 Data source & features

Source (API, no scraping)	Used for	Format · resolution
SMARD (Bundesnetzagentur)	Label: DE-LU day-ahead price (filter 4169); actual solar, wind, load as lag features	JSON, no key · hourly since Oct 2018 · CC BY 4.0
Open-Meteo	Weather forecasts for D: live; history = forecasts as issued 2 days ahead (Previous Runs API)	JSON/CSV, no key · hourly, history from Jan 2024
Fallback: Energy-Charts	Same prices & generation if SMARD is down	JSON, no key · 15-min

Volume: ≈ 24k hourly rows since Jan 2024, full history via API (no poller wait), +24 rows per day.

LABEL

y_neg = DE-LU day-ahead price < 0 for hour h of D. A market outcome published at 12:45.

RARE POSITIVE CLASS (3–6 % OF HOURS)

Only about 1 in 20 hours is negative. Training gives those hours extra weight, the yes/no cut-off is chosen on validation data, and results are measured with metrics made for rare events (PR-AUC, F1), month by month, not with accuracy.

NAMED FEATURES (KNOWN BY 10:00 ON D-1)

Weather fc: irradiance, cloud cover, wind 100 m, temperature, residual-load proxy
Market: price of hour h on D-1 and D-7, negative hours in last 7 days, actual load on D-2
Lags: actual solar & wind on D-2 · **Calendar:** hour, weekday, holiday, month

LEAKAGE: WHAT IS KNOWN WHEN

Time	Event	Usable as feature?
D-2, 12:45	Prices for D-1	Yes, as lag feature
D-1, morning	Latest weather forecast for D	Yes
D-1, 10:00	Cut-off: features pulled, forecast issued	Only data published before this
D-1, 12:45	Prices for D published (= label)	No, never a feature

- **Out-of-time split:** train 2024 · validate H1 2025 · test Jul 2025–Jun 2026 · live from ~Nov 2026.
- Weather from **forecasts issued 2 days ahead**, never observed weather. Every row has an `available_at` timestamp; each row is built only from data published before 10:00 on D-1.

4 System design

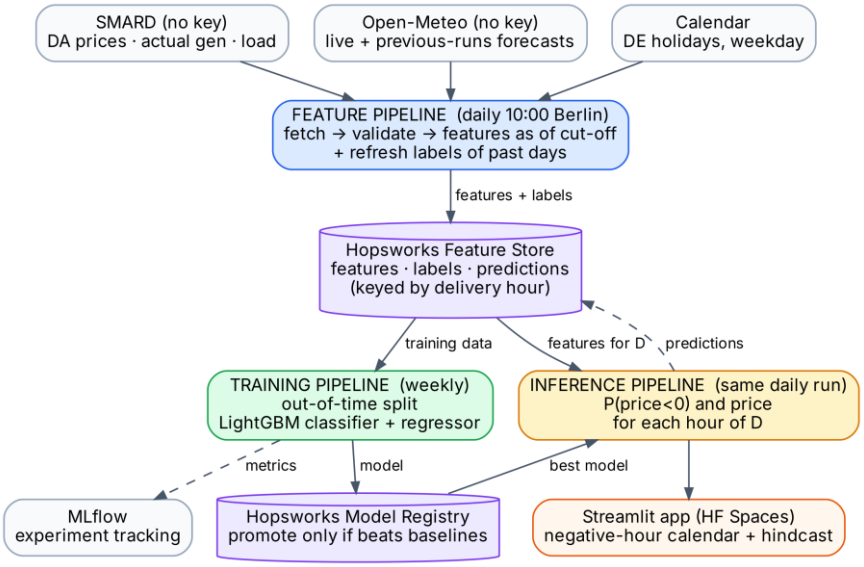


Figure 1 · Feature / Training / Inference pipelines

Layer	Choice	Why
Feature store	Hopsworks	Stores features, labels and predictions per delivery hour; includes the model registry.
Experiment tracking	MLflow (DagsHub)	Every run logged next to the baseline scores.
Orchestration	GitHub Actions cron	Free for public repos. Daily run at 10:00 (features, then inference), 2 h before gate closure; training weekly.
Serving	Batch → Streamlit (HF Spaces)	One forecast a day is enough; app shows tomorrow's risk per hour, then forecast vs. actual after 12:45.